

Week 3 DAY - 11 Hands-On Activities – Keerthivasan R

Problem 1

Problem Statement :

Student Feedback Form Interaction

Scenario:

A training institute wants to collect feedback from students through a simple web form. They want to enhance user experience using jQuery to validate inputs and display messages dynamically without reloading the page.

Requirements

- * Create a feedback form with fields: Name, Email, Course Rating (dropdown), and Comments.
- * Use jQuery selectors to capture form inputs.
- * On clicking the Submit button, validate that Name and Email are not empty.
- * Display a success message dynamically below the form using jQuery DOM manipulation.
- * Clear the form fields after successful submission.

🔍 Technical Constraints

- * Use jQuery CDN for integration.
- * Use basic jQuery syntax: `$(selector).action()`.
- * Do not use plain JavaScript for validation (only jQuery methods).

No backend integration required.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Feedback Form</title>
  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;

      /* Centering */
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
  </style>

```

```
.container {
  width: 400px;
  background: #ffffff;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
  text-align: center;
}

h2 {
  margin-bottom: 20px;
  font-size: 26px;
}

input,
select,
textarea {
  width: 100%;
  padding: 12px;
  margin: 10px 0;
  border-radius: 6px;
  border: 1px solid #ccc;
  font-size: 14px;
  transition: 0.3s ease;
  box-sizing: border-box;
}

input:focus,
select:focus,
textarea:focus {
  border-color: #4a90e2;
  outline: none;
  box-shadow: 0 0 5px rgba(74, 144, 226, 0.4);
}

button {
  width: 100%;
  padding: 12px;
  margin-top: 10px;
  border: none;
  border-radius: 6px;
  background: linear-gradient(to right, #5dade2, #3498db);
  color: white;
  font-size: 16px;
  cursor: pointer;
  transition: 0.3s;
}

button:hover {
  opacity: 0.9;
}

#message {
  margin-top: 15px;
}
```

```

        font-size: 14px;
        text-align: left;
    }

    .error {
        color: red;
    }

    .success {
        color: green;
    }
</style>
</head>

<body>

    <body>

        <div class="container">
            <h2>Student Feedback Form</h2>

            <form id="feedbackform">
                <input type="text" id="name" placeholder="Enter your name">
                <input type="email" id="email" placeholder="Enter your email">

                <select id="rating">
                    <option value="">Select Rating</option>
                    <option value="Excellent">Excellent</option>
                    <option value="Good">Good</option>
                    <option value="Average">Average</option>
                </select>

                <textarea id="comments" placeholder="Enter your comments"></textarea>

                <button type="button" id="submitBtn">Submit Feedback</button>
            </form>

            <div id="message"></div>
        </div>

        <script>
            $(document).ready(function () {

                //Click Event on Submit Button
                $("#submitBtn").click(function () {

                    // Capture input values using jQuery .val()
                    var name = $("#name").val();
                    var email = $("#email").val();
                    var rating = $("#rating").val();
                    var comments = $("#comments").val();

                    //Clear previous messages
                    $("#message").removeClass("error success").text("");

```

```

        //Validation: Check if Name or Email is empty
        if (name.trim() === "" || email.trim() === "" || rating === "" ||
comments.trim() === "") {

            //Show error message dynamically
            $("#message")
                .addClass("error")
                .text("Please fill in all fields.");

        } else {

            //Show success message dynamically
            $("#message")
                .addClass("success")
                .html("Thank you <b>" + name + "</b>! Your feedback has been
submitted successfully.");

            // Clear form fields using jQuery
            $("#feedbackForm")[0].reset();
        }

    });

});
</script>
</body>
</html>

```

OUTPUT:

The image shows a web form titled "Student Feedback Form". It has the following elements:

- Name/Username**: A text input field containing "Keerthi".
- Email**: A text input field containing "keerthi@gmail.com".
- Feedback**: A text area with a dropdown arrow, containing the word "Excellent".
- Rate Product**: A text input field.
- Submit Feedback**: A blue button.
- Message**: A green box at the bottom stating "Thank you, Keerthi! Your feedback has been submitted successfully."

Problem 2

Problem Statement :

A website contains a Frequently Asked Questions (FAQ) section. The company wants answers to appear and disappear when users click on the questions.

Requirements

- Create at least 5 FAQ questions with hidden answers.
- Use jQuery click() event to toggle answers.
- Apply slideToggle() or show()/hide() methods.
- Highlight the selected question using CSS class manipulation.

❑ Technical Constraints

- Use only jQuery methods for DOM manipulation.
- Use class selectors and ID selectors appropriately.
- Avoid inline JavaScript.

Expectations:

- Smooth UI interaction.
- Proper use of jQuery effects.
- Organized HTML structure.

Learning Outcome

- Implement event handling using jQuery.
- Use basic jQuery effects.

Practice DOM traversal and manipulation.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>FAQ Toggle Section</title>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet" href="style.css">
  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      padding: 40px;
    }
  </style>
</head>

<body>
```

```

.faq-container {
    max-width: 700px;
    margin: auto;
}

.faq-item {
    background: #ffffff;
    margin-bottom: 10px;
    border-radius: 5px;
    overflow: hidden;
    border: 1px solid #ddd;
}

.question {
    padding: 15px;
    cursor: pointer;
    font-weight: bold;
    background-color: #f8f9fa;
}

.question.active {
    background-color: #ff1100;
    color: white;
}

.answer {
    display: none;
    padding: 15px;
    border-top: 1px solid #ddd;
    background-color: #ffffff;
}
</style>
</head>
<body>
    <div class="faq-container">
        <h2>Frequently Asked Questions</h2>
        <div class="faq-item">
            <div class="question">1. What services do you offer?</div>
            <div class="answer">I offer web development services including responsive
design and UI development.</div>
        </div>
        <div class="faq-item">
            <div class="question">2. What technologies do you use?</div>
            <div class="answer">I use HTML, CSS, JavaScript, Bootstrap, and jQuery.</div>
        </div>
        <div class="faq-item">
            <div class="question">3. Are your websites mobile friendly?</div>
            <div class="answer">Yes, all websites are fully responsive and optimized for
mobile devices.</div>
        </div>
        <div class="faq-item">
            <div class="question">4. How can I contact you?</div>
            <div class="answer">You can contact me via email or through the contact form
on my website.</div>
        </div>
    </div>

```

```

    <div class="faq-item">
      <div class="question">5. Do you provide maintenance support?</div>
      <div class="answer">Yes, I provide website maintenance and updates as
needed.</div>
    </div>
  </div>
</script>
$(document).ready(function () {

    $(".question").click(function () {

        // Remove active class from other questions
        $(".question").not(this).removeClass("active");

        // Toggle active class for clicked question
        $(this).toggleClass("active");

        // Slide toggle current answer
        $(this).next(".answer").slideToggle();

        // Close other open answers
        $(".answer").not($(this).next()).slideUp();

    });
});
</script>
</body>
</html>

```

OUTPUT:

Frequently Asked Questions

1. What services do you offer?

2. What technologies do you use?

3. Are your websites mobile friendly?

4. How can I contact you?

5. Do you provide maintenance support?

Yes, I provide website maintenance and updates as needed.

Problem 3

Problem Statement :

Problem 3 (Level-2): Product List with Add to Cart Counter

Scenario:

An e-commerce website wants to display a list of products. When users click 'Add to Cart', the cart counter at the top should increase dynamically.

Requirements

- Display at least 4 products with 'Add to Cart' buttons.
- Use jQuery to handle button click events.
- Increase the cart counter dynamically without page refresh.
- Disable the button once the product is added.
- Display a small confirmation message beside the product.

□ Technical Constraints

- Use jQuery event delegation if needed.
- Use .text(), .attr(), and .prop() methods.
- No backend or database integration.

Expectations:

- Efficient event handling.
- Proper use of jQuery methods for attribute manipulation.

Clean UI updates.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Product List with Cart Counter</title>
  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      padding: 30px;
    }
    .header {
      display: flex;
      justify-content: space-between;
      align-items: center;
      margin-bottom: 30px;
    }
  </style>
</head>
<body>
  <div class="header">
    <div class="cart-counter">Cart: 0</div>
    <div class="product-list">
      <div class="product">
        <div class="product-image"><img alt="Product Image" data-bbox="100 100 150 150" /></div>
        <div class="product-info">
          <div class="product-name">Product Name</div>
          <div class="product-price">Product Price</div>
          <div class="add-to-cart">Add to Cart</div>
        </div>
      </div>
      <div class="product">
        <div class="product-image"><img alt="Product Image" data-bbox="100 100 150 150" /></div>
        <div class="product-info">
          <div class="product-name">Product Name</div>
          <div class="product-price">Product Price</div>
          <div class="add-to-cart">Add to Cart</div>
        </div>
      <div class="product">
        <div class="product-image"><img alt="Product Image" data-bbox="100 100 150 150" /></div>
        <div class="product-info">
          <div class="product-name">Product Name</div>
          <div class="product-price">Product Price</div>
          <div class="add-to-cart">Add to Cart</div>
        </div>
      <div class="product">
        <div class="product-image"><img alt="Product Image" data-bbox="100 100 150 150" /></div>
        <div class="product-info">
          <div class="product-name">Product Name</div>
          <div class="product-price">Product Price</div>
          <div class="add-to-cart">Add to Cart</div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

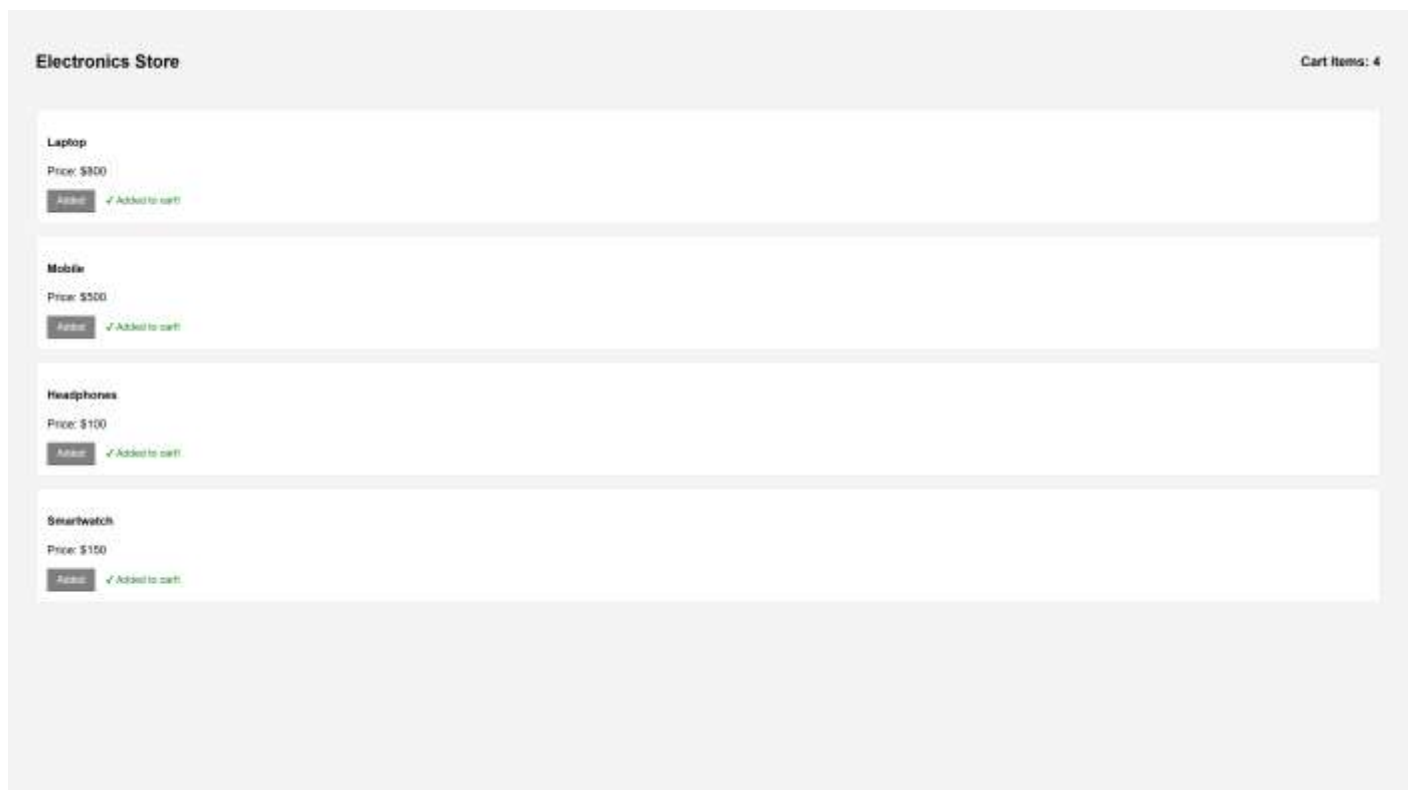


```

        <p>Price: $150</p>
        <button class="add-to-cart" data-product="Smartwatch">Add to Cart</button>
        <span class="added-msg"></span>
    </div>
</div>
<script>
    $(document).ready(function () {
        let cartCount = 0;
        // Event Delegation
        $("#product-list").on("click", ".add-to-cart", function () {
            // Increase counter
            cartCount++;
            $("#cart-count").text(cartCount); // .text() used
            // Disable button
            $(this).prop("disabled", true); // .prop() used
            $(this).addClass("disabled-btn");
            // Change button text
            $(this).text("Added");
            // Store attribute (example usage)
            $(this).attr("data-added", "true"); // .attr() used
            // Show confirmation message
            $(this).siblings(".added-msg").text("✔ Added to cart!");
        });
    });
</script>
</body>
</html>

```

OUTPUT:



Problem 4

Problem Statement :

Interactive To-Do List Application

Scenario:

A productivity web app requires a dynamic To-Do List where users can add, mark as complete, and remove tasks using jQuery.

Requirements

- Provide an input field and Add button to insert tasks.
- Display tasks dynamically in a list.
- Allow users to mark tasks as completed (strike-through effect).
- Provide a delete button for each task.
- Display total task count and completed task count dynamically.

□ Technical Constraints

- Use jQuery for all DOM manipulation and event handling.
- Use append(), remove(), toggleClass(), and on() methods.
- No external libraries except jQuery.

Expectations:

- Proper dynamic element handling.
- Efficient event binding for dynamically created elements.
- Responsive and interactive UI behavior.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Interactive To-Do List</title>

  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      padding: 30px;
      max-width: 600px;
      margin: auto;
    }
  </style>
</head>

<body>
  <div>
    <input type="text" value="Add new task" />
    <button type="button" value="Add" />
  </div>
  <div>
    <ul>
      <li>Task 1</li>
      <li>Task 2</li>
      <li>Task 3</li>
    </ul>
  </div>
  <div>
    <span>Total tasks: 3</span>
    <span>Completed: 0</span>
  </div>
</body>
</html>
```

```
h2 {
  text-align: center;
}

.input-section {
  display: flex;
  gap: 10px;
  margin-bottom: 20px;
}

input {
  flex: 1;
  padding: 8px;
}

button {
  padding: 8px 12px;
  cursor: pointer;
}

ul {
  list-style: none;
  padding: 0;
}

li {
  background: white;
  padding: 10px;
  margin-bottom: 8px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-radius: 5px;
  border: 1px solid #ddd;
}

.completed {
  text-decoration: line-through;
  color: gray;
}

.stats {
  margin-top: 15px;
  font-weight: bold;
}

.delete-btn {
  background: red;
  color: white;
  border: none;
  padding: 5px 8px;
}

</style>
</head>
```

```

<body>

  <h2>My To-Do List</h2>

  <div class="input-section">
    <input type="text" id="taskInput" placeholder="Enter a task">
    <button id="addTask">Add</button>
  </div>

  <ul id="taskList"></ul>

  <div class="stats">
    Total Tasks: <span id="totalCount">0</span> |
    Completed: <span id="completedCount">0</span>
  </div>

  <script>
    $(document).ready(function () {

      function updateCounts() {
        $("#totalCount").text($("#taskList li").length);
        $("#completedCount").text($("#taskList li.completed").length);
      }

      // Add Task
      $("#addTask").on("click", function () {

        let taskText = $("#taskInput").val().trim();

        if (taskText !== "") {

          let newTask = `
          <li>
            <span class="task-text">${taskText}</span>
            <button class="delete-btn">Delete</button>
          </li>
          `;

          $("#taskList").append(newTask); // append() used
          $("#taskInput").val("");
          updateCounts();
        }
      });

      // Mark as Completed (Event Delegation)
      $("#taskList").on("click", ".task-text", function () {
        $(this).parent().toggleClass("completed"); // toggleClass() used
        updateCounts();
      });

      // Delete Task (Event Delegation)
      $("#taskList").on("click", ".delete-btn", function () {
        $(this).parent().remove(); // remove() used
        updateCounts();
      });
    });
  </script>

```

```
});  
</script>  
  
</body>  
  
</html>
```

OUTPUT:

My To-Do List

Design

Develop

Total Tasks: 2 | Completed: 0